Make the tradeoff visible

Compare gains, costs, and who bears them before recommending a choice.

Suggested teaching time: 30 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: None

Students will be able to

- Identify a possible benefit, cost, and affected group.
- Compare feasible allocations of a limited resource.
- Explain how evidence and priorities can change a recommendation.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; no student devices required.
- Instructor guide with model answers, kept separate from the handout.
- Read the three scenarios and their model answers.
- Print the three-page handout or display one activity at a time.
- Explain that all scenarios and figures are invented for reasoning practice.

Teaching sequence

5 min	**Name what is given up.** Use the lunchtime warm-up to distinguish time gained for one activity from time lost for another.
10 min	**Compare study-time allocations.** Pairs complete the school activity. Check that each allocation totals 60 minutes and that expected score gains remain unproven.
8 min	**Discuss priorities and evidence.** Compare the model answer and alternative feasible allocations. Ask learners to name who bears each cost.
7 min	**Explain an independent choice.** Learners complete the community-room exit ticket. Assess the budget, stated priority, tradeoff, and useful question rather than rewarding a particular preference.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / A lunchtime choice

Student task and instructor answer

Explain these words as needed

Tradeoff: A gain in one respect accompanied by a cost or lost opportunity in another.

Opportunity cost: The benefit given up by using a limited resource for something else.

Scenario

Invented scenario: A class has a 40-minute lunch break. One proposal uses 10 minutes for quiet reading and leaves 30 minutes for eating and conversation. The current break allows all 40 minutes for eating and conversation.

Student output: Write one gain, one lost opportunity, and one group that could value that loss.

1. Name the time gained for reading.

2. Name the time given up for another activity.

3. Identify students who could especially value the activity losing time.

Student success check: Have you described both sides without assuming everyone has the same priority?

Model answer

The proposal adds 10 minutes of reading and removes 10 minutes available for eating or conversation. Students who need a longer eating break or value lunchtime social contact could especially value the lost time.

Why this meets the task: The answer identifies the fixed time budget and a possible affected group without deciding that one use is best for everyone.

Main activity / How should study time be used?

Student task and instructor answer

Scenario

Invented scenario: A school has 60 minutes each week for optional study activities. It proposes moving 30 minutes from open reading to test preparation. Teachers expect more exam practice, but no evidence yet shows a score gain. Students who rely on school reading time would lose half that time.

Student output: Write a benefit, cost, and affected group; propose a feasible allocation; add one question to guide the decision.

1. Make three labelled lines: possible benefit, cost, and affected group. Use the facts and distinguish an expected benefit from a demonstrated result.
2. Propose one alternative allocation of the 60 minutes and explain its tradeoff.
3. Write one question the school should answer before choosing a plan.

Student success check: Do all allocations total 60 minutes? Have you named who bears the cost rather than calling the proposal simply 'good' or 'bad'?

Model answer

Possible benefit: more exam practice, with score improvement still unproven. Cost: 30 fewer minutes of open reading. Affected group: especially students dependent on school reading time. An alternative is 15 minutes of preparation and 45 of reading; it preserves more reading but provides less preparation. Ask whether a trial shows gains large enough to justify the lost reading opportunity.

Why this meets the task: It makes the limited resource explicit and compares benefits, costs, and affected students without inventing guaranteed gains.

Exit ticket / Sharing the community room

Student task and instructor answer

Scenario

Invented scenario: A community center has six evening room-hours to allocate each week. It currently offers six hours of homework help. A proposal would use two hours for art and four for homework help. Some students have no quiet study space at home. No results for the art program are available.

Student output: Write a benefit, cost, and affected group; recommend a six-hour allocation with a reason; ask one useful question.

1. Identify a possible gain from art and the loss from reducing homework help, including who may be affected.
2. Choose hours for art and homework help totaling six, and explain your priority and its cost.
3. Name one question about outcomes or needs that could change your recommendation.

Student success check: Does your allocation total six, and have you acknowledged the cost of your preferred option?

Model answer

Art offers an additional creative activity; reducing homework help removes two supported study hours, affecting students without quiet space. One defensible plan is one hour of art and five of homework help, prioritizing study access while allowing some art; it offers less art time than the proposal. Ask how many students depend on each homework-help hour and what benefits an art trial shows. Other feasible allocations can be justified with different priorities.

Why this meets the task: The answer makes the limited resource, affected students, and priority explicit without claiming a uniquely correct allocation or invented program results.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- Benefits, costs, and an affected group are identified without inventing outcomes.
- The proposed allocation fits the stated resource limit and explains its priority and cost.
- The follow-up question could inform the decision while allowing different reasonable priorities.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Can two reasonable recommendations use the same facts?

Listen for: Yes. People may give different weight to study access, creativity, or other aims. Each should state its priority, acknowledge costs, and use the facts accurately.

What is known about the test-preparation benefit?

Listen for: More practice time is expected, but improved scores have not been demonstrated. Keep the immediate allocation separate from the hoped-for result.

Common confusions

A benefit makes a proposal best overall. Compare the gain with costs, lost opportunities, and who is affected.

A prediction is already a demonstrated result. Label expected benefits as possibilities until suitable evidence supports them.

There must be one correct allocation. Different priorities may support different feasible plans. Evaluate the reasoning and evidence, not agreement with one preference.

Adapt the lesson

More support: Draw a 60-minute bar for the main task. Label preparation and reading, then use the headings gain, loss, affected group. Expected example: 15 minutes for preparation and 45 for reading preserves more reading but supplies less preparation than the proposal.

More challenge: Ask whether dividing time equally is automatically fairest. Expected response: no; equal time is one rule, but needs, benefits, and access may justify another allocation. Any proposed rule should state its purpose and costs.

Next practice

https://xixteen.com/learners/?practice=tradeoffs#workbench